



RoadSide

Your motorcycle tells you when something is wrong.
RoadSide is the app that actually listens.

Offline, on-device audio + vision diagnostics for two-wheelers

THE PROBLEM

A strange noise, 40 km from the nearest mechanic

1 You can hear it

Riders notice a new rattle, squeal or click long before a breakdown. They just cannot name it.

2 You cannot search it

Describing a sound in a search box does not work, and highway coverage in India is patchy at best.

3 You cannot risk it

Ignore a dry chain and it snaps. Over-service it and you pay for work the bike never needed.

“

There is a ticking sound when I pull away. Is it the chain, the engine, or nothing at all?

Today the only answer is to ride to a mechanic and hope.

THE PRODUCT

Record. Point the camera. Get plain-language evidence.

1

Listen

Hold the phone near the bike for five seconds. RoadSide records at 16 kHz and classifies what it hears.

2

Look

Point the camera at the chain.
RoadSide locates the drivetrain in the frame and reports what it can see.

3

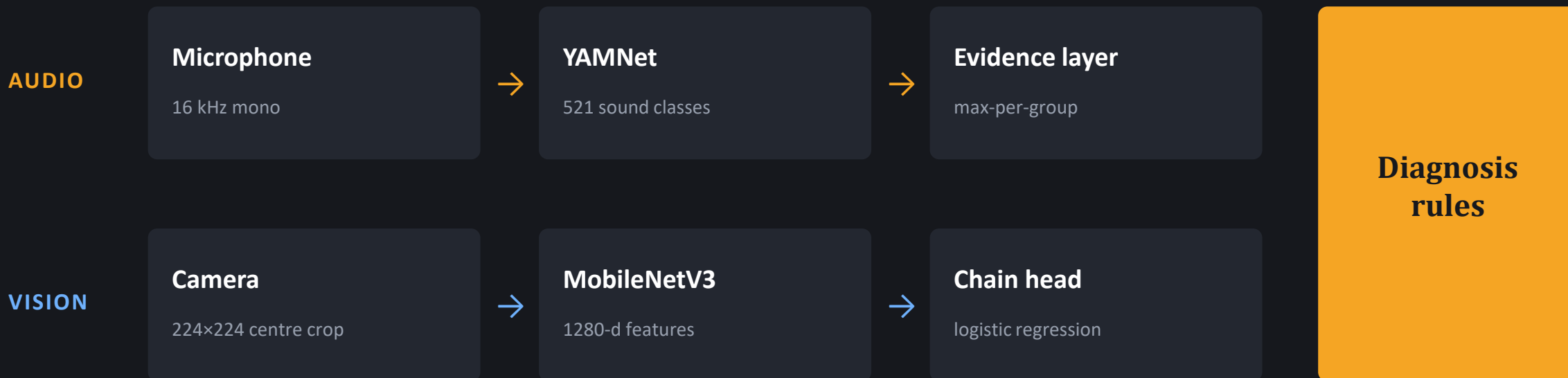
Decide

Sound and sight are fused into one advisory, with the likely cause, the tools needed and a safety note.

No account. No upload. No signal required — the app has no internet permission at all.

HOW IT WORKS

Two pretrained backbones, two tiny heads, one rules engine



Both backbones are frozen and pretrained. We train only the small heads, so the whole model set is 35 MB and fits in an APK.

NOT A MOCKUP

Measured on a real OnePlus 13R, not estimated

14 ms

Audio model cold start

2 ms

Per-window inference

43 ms

Full 5-second clip

30 ms

Per image, vision

14 / 14 instrumented tests pass on device

- Model load, inference, preprocessing, evidence rules, fusion logic and crash checks — run headlessly over ADB
- Airplane-mode pass: audio and vision both verified with every radio off
- Zero crashes, zero ANRs, 16 KB page-size compliant for future Android kernels

**One command
reproduces it all**

`verify.ps1`

It reports evidence, and says “I don’t know”

Most demo AI guesses confidently. A wrong confident answer about a brake is worse than no answer.

● What we refused to ship

- A rust detector keyed on brown backgrounds — we measured it and clean chains on wooden benches scored higher than rusty ones
- A chain-condition class trained on 3 images
- Any claim that a passing motorcycle is a healthy one

● What we shipped instead

- Separate acoustic evidence from mechanical verdict
- UNKNOWN whenever confidence or margin is too low
- Every threshold documented as provisional until real data arrives

Across 98 real traffic recordings, zero motorcycles were falsely flagged as chain or brake faults.

VALIDATION

Tested against public datasets, not our own recordings

Audio · IDMT-Traffic

90 clips pulled from a 9.7 GB Fraunhofer dataset, plus 8 openly licensed close-mic recordings: 58 motorcycles, 20 background, 20 other vehicles.

0 / 58

motorcycles wrongly flagged as a chain fault

0 / 20

background clips flagged as mechanical

Vision · Wikimedia Commons

51 openly licensed chain and non-chain photographs, each labelled by eye rather than trusting the category name.

76.5 %

chain detection, cross-validated

Cross-validated means every prediction came from a fold that never saw that image. We report that number, not the flattering one.

The breakdown happens where the signal isn't

1 Works with no network

The app declares no internet permission at all — it is structurally incapable of phoning home. Verified in airplane mode.

2 Costs nothing to run

No inference bills, no API keys, no per-user cost. A 35 MB model set runs on the handset the rider already owns.

3 Private by construction

Engine audio and vehicle photos are processed locally and are not uploaded by the app.

4 Instant

43 ms for a five-second clip. The answer is there before the rider has put the phone down.

RoadSide works with the radios off, which is exactly where a breakdown happens.

WHAT'S NEXT

The pipeline is built. It is waiting on labelled data.

NOW

Working prototype

Audio evidence, chain detection and fusion running offline on device, with 1024-d YAMNet embeddings already extracted and ready.

NEXT

Labelled fault data

A service-centre partnership to record chains, brakes and engines in known good and known bad states. No open dataset contains this.

THEN

Specialist classifier

A small head over the embeddings: NORMAL / CHAIN / BRAKE / ENGINE. The extraction pipeline is done; only the labels are missing.

India has over 220 million two-wheelers, and almost none of them come with a diagnostic tool.



RoadSide

A mechanic's ear, in every rider's pocket — with no signal, no subscription and no guesswork.

Fully offline

Verified on device

Evidence, not guesses